Classical Theoretical Foundations of Missing Data Inferences

This report compiles the pre-2018 classical statistical foundations that underwrite modern deep learning imputation architectures. When modern frameworks (such as *not-MIWAE*, *PSMVAE*, or *FragmGAN*) handle non-ignorable missingness (MNAR), they directly translate, scale, or parameterize these exact mathematical structures.

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1. Selection Models vs. Pattern-Mixture Models

To model the joint distribution of the data matrix Y and the missingness indicator mask M, classical statistics established two distinct factorizations. Modern deep learning models are fundamentally divided along these same two conceptual lines.

Heckman, J. J. (1979)

- **Title:** *"Sample selection bias as a specification error"*
- **Theoretical Contribution:** Formalizes the classical **Selection Model** factorization:

  $$f(Y, M \mid X) = f(Y \mid X)P(M \mid Y, X)$$

  It demonstrates that when missingness is non-random, treating the observed data as a representative sample introduces a specification bias analogous to omitted variables.
- **Modern Bridge:** Direct theoretical ancestor to **not-MIWAE** (Ipsen et al., 2021) and **GNR** (Chen et al., 2023), which use deep neural networks to model the highly complex conditional distribution of the missingness mask $P(M \mid Y)$ rather than assuming ignorability.

Little, R. J. (1993)

- **Title:** *"Pattern-mixture models for multivariate incomplete data"*
- **Theoretical Contribution:** Formalizes the **Pattern-Mixture Model** factorization:

  $$f(Y, M \mid X) = f(Y \mid M, X)P(M \mid X)$$

  This approach models distinct, pattern-specific distributions for the data conditioned on the observed missingness pattern, utilizing identifying restrictions to handle the unobserved strata.
- **Modern Bridge:** Directly underpins **PSMVAE** (Ghalebikesabi et al., 2021), which maps these classical pattern-specific mixture distributions into deep latent spaces using variational autoencoders.

Diggle, P. J., & Kenward, M. G. (1994)

- **Title:** *"Informative drop-out in longitudinal data analysis"*
- **Theoretical Contribution:** Establishes the first fully parametric longitudinal selection model for continuous data with informative dropout.
- **Modern Bridge:** Serves as the mathematical benchmark for modern recurrent and bidirectional MNAR imputation networks (e.g., BRITS) that attempt to capture temporal decay and informative transitions.

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2. The Taxonomy of MNAR

Before a deep learning model can be engineered to process non-randomly missing data, researchers must characterize *how* the missingness process behaves. The formal taxonomy of these mechanisms originates in these works.

Little, R. J. (1995)

- **Title:** *"Modeling the drop-out mechanism in repeated-measures studies"*
- **Theoretical Contribution:** Systematically defines and contrasts covariate-dependent dropout (where missingness depends on observed baseline characteristics) and outcome-dependent dropout (self-masking MNAR).
- **Modern Bridge:** Establishes the classification system and testing criteria used to design synthetic missingness masks in modern deep learning benchmarks (such as the *Beyond Accuracy* 2025 study).

Heckman, J. J. (1976)

- **Title:** *"The common structure of statistical models of truncation, sample selection and limited dependent variables and a simple estimator for such models"*
- **Theoretical Contribution:** Mathematically defines threshold-based truncation and self-censoring, establishing the limits of what can be recovered when data is systematically missing due to its own extreme values.
- **Modern Bridge:** Essential for understanding **sensor clipping and physical saturation** in environmental monitoring (e.g., air pollution monitors failing precisely when particulate concentrations exceed toxic thresholds).

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3. Sensitivity Analysis & Tipping-Point Origins

Because MNAR mechanisms cannot be verified using observed data alone, classical statistics pioneered sensitivity analysis. Rather than searching for a single "perfect" imputation, this paradigm evaluates how robust conclusions are to variations in non-random assumptions.

Scharfstein, D. O., Rotnitzky, A., & Robins, J. M. (1999)

- **Title:** *"Adjusting for nonignorable consecutive drop-out in semiparametric nonresponse models"*
- **Theoretical Contribution:** Introduces semiparametric sensitivity parameters to systematically perturb the probability of non-response, validating the use of non-identifiable offsets to map boundaries of statistical inference.
- **Modern Bridge:** Groundwork for the mathematical integration of structured sensitivity bounds into generative autoencoders and probabilistic predictors.

Carpenter, J. R., Kenward, M. G., & White, I. R. (2007)

- **Title:** *"Sensitivity analysis after multiple imputation under missing at random: a weighting approach"*
- **Theoretical Contribution:** Introduces the formal δ**-adjustment (offset) method** within multiple imputation frameworks, where imputed values are systematically shifted by a sensitivity parameter δ.
- **Modern Bridge:** This direct mathematical formulation is what multivariable sensitivity frameworks—such as **NARFCS** (Tompsett et al., 2018)—generalize and scale for high-dimensional settings.

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4. Ignorability & Its Consequences

To critically evaluate what modern deep learning models gain—and what they risk losing—one must return to the foundational theorems defining the boundaries of statistical ignorability.

Rubin, D. B. (1976)

- **Title:** *"Inference and missing data"*
- **Theoretical Contribution:** The seminal mathematical formulation of Missing Completely at Random (MCAR), Missing at Random (MAR), and the exact conditions under which the missingness mechanism is "ignorable" for likelihood-based and Bayesian inferences.
- **Modern Bridge:** The ultimate yardstick of the field. Every modern deep learning paper makes implicit or explicit assumptions regarding Rubin's taxonomy (e.g., *FragmGAN* proving that GAIN's hint matrix restricts its validity to Rubin's MCAR condition).

Little, R. J., & Rubin, D. B. (1987 / 2002)

- **Title:** *Statistical Analysis with Missing Data (Textbook)*
- **Theoretical Contribution:** The definitive textbook of the discipline. It mathematically demonstrates that applying MAR-assuming imputation methods (like standard expectation-maximization or basic multiple imputation) to MNAR data introduces systematic asymptotic bias and severely invalidates downstream hypothesis testing.
- **Modern Bridge:** Essential reading for the "Inference vs. Reconstruction" debate, proving why minimizing point-accuracy error (lowest RMSE) often comes at the cost of statistical validity.

Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977)

- **Title:** *"Maximum Likelihood from Incomplete Data via the EM Algorithm"*
- **Theoretical Contribution:** Formalizes the Expectation-Maximization (EM) algorithm for computing maximum likelihood estimates from incomplete data under MAR.
- **Modern Bridge:** The classic mathematical optimization engine that modern latent-variable deep architectures (such as VAEs, MIWAE, and diffusion models) seek to parallel and accelerate in high dimensions.